(i)

$$\because \frac{d}{dx}\left(2x\sqrt{1+x^3}\right) = 2\sqrt{1+x^3} + 2x \cdot \frac{3x^2}{2\sqrt{1+x^3}}$$

$$\therefore \int \left(2\sqrt{1+x^3} + \frac{3x^3}{\sqrt{1+x^3}}\right) dx = 2x\sqrt{1+x^3} + C$$

(ii)

$$(x^2+2)\frac{\sin x}{x^3} = \frac{\sin x}{x} + \frac{2\sin x}{x^3}$$

$$\int \frac{2\sin x}{x^3}\,dx = -\frac{\sin x}{x^2} + \int \frac{\cos x}{x^2}\,dx$$

$$\int \frac{\sin x}{x}\,dx = -\frac{\cos x}{x} - \int \frac{\cos x}{x^2}\,dx$$

$$\int (x^2+2)\frac{\sin x}{x^3}\,dx = \int \frac{\sin x}{x}\,dx - \frac{\sin x}{x^2} - \frac{\cos x}{x} - \int \frac{\sin x}{x}\,dx$$

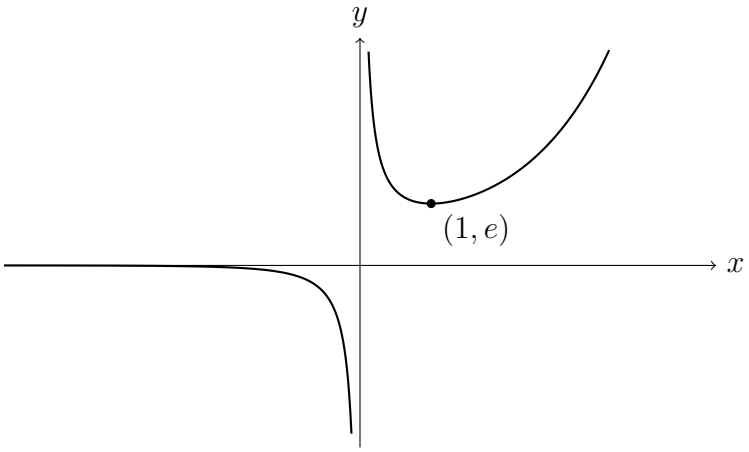
$$= -\frac{\sin x + x\cos x}{x^2} + C$$

(iii)(a)

$$\frac{dy}{dx} = \frac{e^x(x-1)}{x^2}$$

$$\frac{dy}{dx} = 0 \implies x = 1$$

x	$-\infty$	0^-	0^+	1	∞
y	0^-	$-\infty$	∞	e	∞



(iii)(b)

$$\int_a^{2a} \frac{e^x}{x}\,dx = \int_a^{2a} \frac{e^x}{x^2}\,dx$$

$$\because \frac{d}{dx}\left(\frac{e^x}{x}\right) = \frac{e^x}{x} - \frac{e^x}{x^2}$$

$$\left[\frac{e^x}{x}\right]_a^{2a} = 0$$

$$\frac{e^{2a}}{2a} - \frac{e^a}{a} = 0$$

$$a = \ln 2$$

(iii)(c)

Assume distinct integers m, n exist.

$$\int_m^n \frac{e^x}{x}\,dx = \int_m^n \frac{e^x}{x^2}\,dx$$

$$\int_m^n \left(\frac{e^x}{x} - \frac{e^x}{x^2}\right)\,dx = 0$$

$$\left[\frac{e^x}{x}\right]_m^n = 0$$

$$\frac{e^n}{n} = \frac{e^m}{m}$$

From **(iii)(a)**, $\dfrac{e^x}{x}$ is strictly decreasing on $x < 0$ and strictly increasing on $x \geq 1$.
If $m, n > 0$,

$$\frac{e^n}{n} = \frac{e^m}{m} \implies m = n$$

If $m, n < 0$,

$$\frac{e^n}{n} = \frac{e^m}{m} \implies m = n$$

If one is negative and one is positive, the integral crosses $x = 0$.

$$\therefore \text{no distinct integers } m, n \text{ exist}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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